

Nawat Pimolsathien

นาวัต พิมลเสถียร

Machine Learning & Software Engineer · Computer Engineering & Digital Technology, Chulalongkorn University

nawatpim@gmail.com · 088-864-5743 · nawatpim.com · github.com/nawat-john · linkedin.com/in/nawat-pimolsathien · Nonthaburi, Thailand

Fourth-year Computer Engineering student at Chulalongkorn University, currently building LLM-based systems at Muang Thai Life Assurance. I build machine learning end to end — data pipelines, model training (QLoRA fine-tuning, pruning, quantization, classical classifiers, reinforcement learning), evaluation and deployment — and I measure results rather than assume them. Alongside that I ship backend software in Python, .NET and Node.js.

FLAGSHIP PROJECT

pholama — compressing and specializing a Thai LLM

github.com/nawat-john/pholama

Pruned and quantized Typhoon2 3B (Llama 3.2 base) into a domain Q&A model for Chulalongkorn's registrar office: vocabulary trimmed 128k→45k, layers pruned 28→24, quality recovered with QLoRA (perplexity 14→7.6), then quantized to GGUF Q4_K_M and deployed behind retrieval. Against the original: -20.5% parameters (3.21B→2.55B), -76% deploy size (6.4 GB→1.5 GB), 23.5 tok/s on CPU against 10.6 tok/s on GPU, and in-domain quality 35.0%→47.5% on a held-out 40-question domain set, LLM-judged with the same retrieval for both. Every stage ran on a 4 GB laptop GPU.

EXPERIENCE

Software Engineer Intern

Aug 2025 — Present

Muang Thai Life Assurance PCL.

- Built a sales-assistance chatbot on a multi-agent + RAG architecture using Python, LangChain and LangGraph, routed by a TF-IDF + logistic-regression intent classifier (scikit-learn).
- Evaluated retrieval with hit-rate and MRR against reference answers generated by a larger model, combined with human scoring.
- Modernize legacy web applications, with unit and integration tests to preserve behavior through the rewrite.

Software Developer (Intern / Part-time)

May 2025 — Sep 2025

Power Partner Co., Ltd.

- Developed CAD add-ins in C# and .NET from real client requirements, and migrated them from .NET Framework to .NET Core LTS.

Teaching Assistant

Aug 2024 — Mar 2026

Chulalongkorn University

- TA for Frontend Development (Next.js), Data Structures & Algorithms, and Digital Computer Logic.
- Debugged student code daily, tracing root causes across frontend and algorithm assignments; graded work and answered technical questions.
- Invited back as Frontend Development TA for a second term.

Software Engineer Intern

May 2024 — Jul 2024

NITMX Co., Ltd.

- Built full-stack websites with Next.js and Node.js, and designed the database schema and UI (Figma).

EDUCATION

B.Eng. Computer Engineering & Digital Technology

2023 — 2026

Chulalongkorn University · GPAX 3.71 / 4.00

Coursework: Introduction to Data Science Engineering (A) · Neural Networks (A) · AI & Machine Learning · Statistics for Computer Engineering · Computer Engineering Mathematics I–II · Software Engineering · Database Systems · Data Structures & Algorithms · Capstone Project I–II (A).

Full transcript and TOEIC score report available at nawatpim.com.

LANGUAGES

Thai — native · English — TOEIC 900 / 990

TOOLKIT

// ai & ml
Python · PyTorch · Hugging Face
Transformers · PEFT / QLoRA · TensorFlow
/ Keras · scikit-learn · PyThaiNLP ·
Gymnasium (RL) · LangChain / LangGraph

// data
pandas · NumPy · SQL · PostgreSQL

// software
.NET / C# · Node.js / NestJS · Next.js ·
React · Docker · Redis

SELECTED PROJECTS

typhoon_finetune

QLoRA fine-tune of Typhoon2-3B on a Thai speech-style dataset, trained inside 4 GB of VRAM (4-bit base, gradient checkpointing, paged 8-bit optimizer), with a data pipeline built from YouTube transcripts.

MooGPT

GPT-style transformer written from scratch in C++ (tensors, backprop, AdamW, BPE tokenizer), every gradient verified against a PyTorch oracle.

air-hockey RL

Hybrid physics and reinforcement-learning agent (self-play) deciding at 10 Hz.

Aurum

private client work
Order management platform for a gold retailer: Next.js + NestJS, PostgreSQL, deployable on-premise or in the cloud. Code private; walkthrough on request.

satang

Event-driven double-entry ledger in Go with a Kafka outbox and OpenTelemetry tracing.

Let's build something

Open to machine learning and software engineering roles.

nawatpim@gmail.com

088-864-5743